

# **Year Up United Data Analyst Academy**

## **Pre-Work Week 2 Lab Workbook**

This courseware is to be distributed to accompany instructor led training.

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# **Module 1**

## **Descriptive Statistics in Excel**

# Exercise 1.A Calculating Descriptive Statistics

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## Scenario

You are reviewing monthly sales data for 12 different retail locations.

Your goal is to understand overall performance using summary statistics.

## Dataset

Create a new spreadsheet named `homework1.xlsx` and save it to your `DataAnalytics/pre-week-2` folder.

Create a worksheet named `Store_Data` and enter:

| Store   | Monthly Sales |
|---------|---------------|
| Store A | 51            |
| Store B | 44            |
| Store C | 49            |
| Store D | 46            |
| Store E | 52            |
| Store F | 47            |
| Store G | 45            |
| Store H | 72            |
| Store I | 43            |
| Store J | 48            |
| Store K | 46            |
| Store L | 50            |

## Tasks

### 1. First calculate:

a) Mean: `=AVERAGE (range)`

b) Median: `=MEDIAN (range)`

c) Mode: `=MODE.SNGL (range)`

d) Range: `MAX (range) - MIN (range)`

### 2. Identify whether any values appear unusually high

## Exercise 1.B Understanding Variability

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### Scenario

Leadership wants to know how consistent store performance is across locations.

### Tasks

1. Calculate the standard deviation: `=STDEV.S(range)`
2. Temporary remove Store H from your calculation by adjusting the cell range in your formula
3. Compare the two results

# Exercise 1.C Interpreting the Results

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## Scenario

You must provide a short written explanation of performance.

## Interpreting Sales Data

1. Create a worksheet named `Store_Summary` and write 5-7 sentences explaining:
  - a) The overall average performance
  - b) Whether store performance is tightly grouped or spread out
  - c) Whether one value significantly influences the results
  - d) What additional information you would want

## Submitting Your Work

1. Ensure you have saved your work as `homework1.xlsx` in your `DataAnalytics/pre-week-2` folder
2. Add, commit, and push your file to GitHub
3. Submit your GitHub repo URL to Brightspace under **Pre-Work Week 2 Homework 1**

# **Module 2**

## **Data Classification & Visualization with Excel**



## Exercise 2.A Data Classification and Sorting

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### Scenario

You are preparing data for a leadership dashboard that will highlight top-performing and underperforming stores.

Before building visuals, you need to organize the data to quickly identify performance differences.

You are now given additional information about each store.

### Dataset

Create a new spreadsheet named `homework2.xlsx` and save it to your `DataAnalytics/pre-week-2` folder.

Create a worksheet named `Store_Data` and enter:

| Store   | Region | Category    | Monthly Sales |
|---------|--------|-------------|---------------|
| Store A | North  | Apparel     | 51            |
| Store B | North  | Electronics | 44            |
| Store C | South  | Apparel     | 49            |
| Store D | South  | Home Goods  | 46            |
| Store E | East   | Electronics | 52            |
| Store F | East   | Apparel     | 47            |
| Store G | West   | Home Goods  | 45            |
| Store H | West   | Electronics | 72            |
| Store I | North  | Home Goods  | 43            |
| Store J | South  | Apparel     | 48            |
| Store K | East   | Home Goods  | 46            |
| Store L | West   | Electronics | 50            |

## Tasks

1. Identify:
  - a) Numerical column
  - b) Categorical columns
2. Select the entire dataset including headers
3. Go to the **Data** tab
4. Click **Sort**
5. Choose Monthly Sales as the column
6. Select Largest to Smallest
7. Identify which store stands out

Observe how performance varies across stores and consider whether one store is significantly higher than the rest.

## Exercise 2.B Regional Summary and Pie Chart

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### Scenario

Leadership is preparing for a quarterly review and wants to understand how total sales are distributed across regions.

Your task is to summarize the data at the regional level and create a visualization that clearly shows how sales are distributed.

### Tasks

1. Create a PivotTable summarizing total Monthly Sales by Region
  - a) Select your full dataset including headers
  - b) Go to the **Insert** tab
  - c) Click **Tables > PivotTable**
  - d) Choose **From Table/Range**
  - e) Place Region in **Rows**
  - f) Place Monthly Sales in **Values**
2. Insert a pie chart based on that summary
  - a) Select the PivotTable summary
  - b) Go to the **Insert** tab
  - c) Choose Pie Chart
3. Format the chart clearly
4. Create a worksheet named `Store_Region_Summary` and write 3-4 sentences explaining:
  - a) Which region contributes the most
  - b) Whether one store may be influencing the totals

## Exercise 2.C Category Comparison and Chart Selection

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### Scenario

The merchandising team wants to compare performance by product category.

### Tasks

1. Create a PivotTable summarizing Monthly Sales by Category
  - a) Select the dataset
  - b) Insert a PivotTable
  - c) Place Category in **Rows**
  - d) Place Monthly Sales in **Values**
  - e) Ensure Values is set to **Sum**
2. Choose an appropriate chart type to compare categories
3. Create a worksheet named `Store_Category_Summary` and write 3-4 sentences explaining:
  - a) Which category performs best
  - b) Why you selected that chart type
  - c) What insight the chart makes clear

### Submitting Your Work

1. Ensure you have saved your work as `homework2.xlsx` in your `DataAnalytics/pre-week-2` folder
2. Add, commit, and push your changes to GitHub
3. Submit your GitHub repo URL to Brightspace under **Pre-Work Week 2 Homework 2**